



BEST PRACTICE SERIES

Continuous Improvement
(Strategic)

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

Continuous Improvement (Strategic)0

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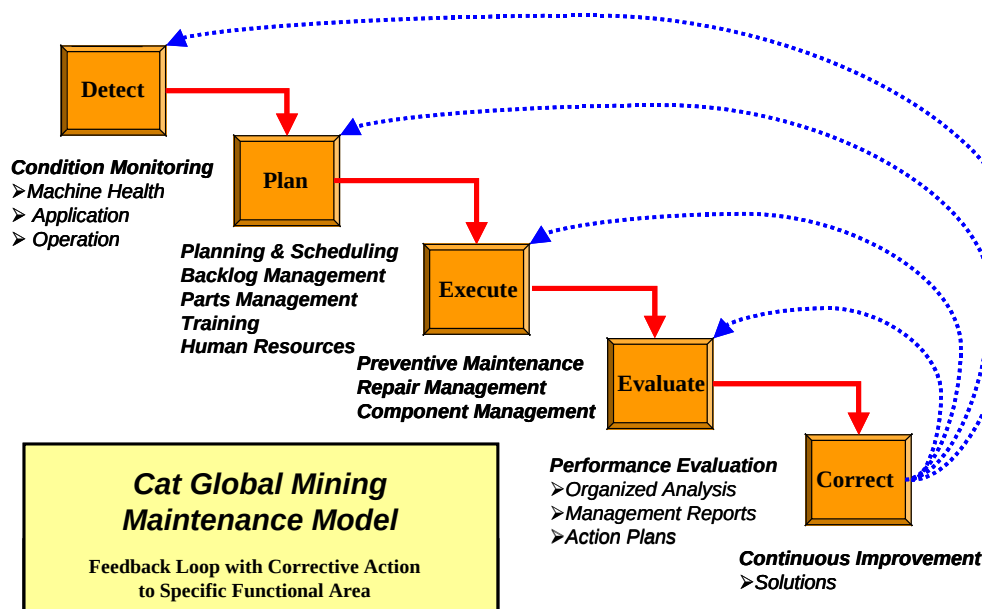
1.0 Introduction

Continuous Improvement is a critical part of the overall Mining Equipment Management process. To provide consistent product support service to mining customers, it is necessary to continuously evaluate all aspects of the operations to identify the gaps in operations and areas that can be improved. This will help you better manage the availability, reliability and cost of the operation.

To accomplish this, a Continuous Improvement program should be implemented that adopts an organized approach to define and analyze problems, prioritize problems based on time, money & resources, determine corrective action and apply the solutions. This approach will need to involve the entire organization, as the input for Continuous Improvement comes from many functional areas within the operation.

The Continuous Improvement process is a loop of an “input of information” that is used in the improvement process to identify gaps in performance, develop a fix and the “feedback of corrective actions” to the various functional areas for implementation. The Continuous Improvement process starts with

- Input from the results of the current actions in the form of KPI's (top tier metrics).
Example: Low Availability
- Drill down into the Process Metrics to find the Root Cause of the poor performance.
Example: MTTR is too high...time to repair.
- Develop Corrective Action(s), example: Provide training for mechanics to improve efficiency.
- Communicate Corrective Action(s) to appropriate areas in organization.



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2.0 Best Practice Description

The first step is to identify the problems using input from the results of current actions in the form of Top Tier Metrics – KPI"s. The Continuous Improvement process should not be limited to technical / "iron" - related issues alone since problems may be the result of operational, service or other product support-related areas, e.g. parts. Problem reports should be analyzed on the basis of machine, system, component, part number and / or frequency with the key criteria for inclusion on the list being costs and downtime.

Once the problems are defined, the challenge for the equipment manager is to identify the root cause(s) of each of the problems and to take action to manage them (minimize their impact on the operation) until they can be resolved. To properly identify the root cause, it will be necessary to drill down into the Top Tier Metrics to identify the process metric that is associated with the poor performance. Cause & Effect Diagrams (Fishbone) are effective tools to identify root cause.

After the root cause has been identified, a solution to the problem can be developed and communicated back to the appropriate area(s) for implementation.

A follow-up check must be performed to validate that the proposed corrective action is producing the desired results.

Ideally, the Continuous Improvement exercise results in problem avoidance or, at least, minimization. Steps toward solution may involve modifications to the machine design or configuration, modification to the operation or application of the machine, modifications to machine maintenance practices, additional manpower, training, tooling or other support resources.

This Best Practice describes the strategic best practice of implementing a Continuous Improvement program.

To have a successful Continuous Improvement program you will need to include the following elements:

- Strategy
- Procedures
- Resources
- Execution/Implementation
- Performance Evaluation

2.1 Strategy

The strategy defines the scope, goals & objectives of your Continuous Improvement Program and should address the following areas:

- "Top Tier" site performance
- Maintenance & Repair management process performance
- Application severity
- Operational issues & operating practices
- Design / technical issues
- Costs / financial performance
- Safety

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- Parts support
- CRC / dealer rebuild capabilities

2.2 Procedures

A Continuous Improvement program will require the use some type of Business Process Improvement methodology that provides the necessary steps to follow and tools to use during the various phases of a project. Some examples of Business Process Improvement methodologies would be 6 Sigma and GMP+.

Part of the business process improvement methodology involves:

- Project identification
- Project prioritization & selection
- Project management

2.3 Resources

A Continuous Improvement team coordinator should be identified to provide leadership to the Continuous Improvement efforts within the organization. This coordinator will provide leadership to a team of subject matter experts (SME's) that will be assigned to each project from the various functional areas depending on the nature of the project. The entire team must be trained and competent in the Business Process Improvement methodologies.

2.4 Execution/Implementation

To ensure successful execution of Continuous Improvement projects, the teams must meet on a regular basis to provide consistency and follow-up on the established work plans. To make sure that the project stays on track, a 3W work plan template should be used to identify:

- **What:** Task or action to be performed
- **Who:** Person responsible for completing the task
- **When:** Date that the task is expected to be completed

What	Who	When
Task 1	Person 1	Date 1
Task 2	Person 2	Date 2
Task 3	Person 3	Date 3
Task 4	Person 4	Date 4

Once the corrective action has been developed and communicated, the appropriate area personnel must implement the fix that has been recommended by the Continuous Improvement team.

2.5 Performance Evaluation

For a fully functional Continuous Improvement program, performance must be measured against pre-established Top Tier Metrics (KPI's) and reported on a regular basis.

Some examples of KPI's for Continuous Improvement are:

- Quantity of Continuous Improvement projects opened in the period

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- Quantity of Continuous Improvement projects active
- Quantity of Continuous Improvement projects closed in the period
- Average project execution time
- Completed project delivered value

These KPI reports should be distributed to the appropriate people in your organization on a monthly basis. The reporting process should include:

- Content of report
- Distribution list
- Frequency of distribution

3.0 Implementation Steps

The Continuous Improvement process consists of the following steps and tools:

- Identifying key problems that are affecting the performance
 - o Comparing actual vs. planned results using KPI's.
 - o Problem reports
 - o Root Cause Analysis (Fishbone Analysis)
 - o Site/Operation Audits
 - Site Assessment Tool
 - Contamination Control Compliance Guide
 - Gap Analysis Tool
- Quantify the impact on the fleet operation
 - o Cost
 - o Availability
 - o Repair Frequency
 - o Repair Duration
 - o Manpower & tooling Requirements
- Prioritizing the problems for selection of candidate Continuous Improvement Projects
 - o A filtering process to evaluate the cost & ease of implementation
- Identification of root cause by drilling down into the process metrics to determine the element that is causing the poor performance.
 - o Cause & Effect Diagrams (Fishbone)
- Identify potential solutions & determine best course of action
 - o Pugh Matrix evaluation
- Develop an action plan which will define the following:
 - o What is the corrective action?
 - o Who is responsible for implementation of this action?
 - o When will this corrective action be completed?
- Implement and complete the plan
- Monitor and quantify the results to verify the effects of the corrective action.
 - o Establish & monitor Continuous Improvement KPI's

Each of these steps should be followed using Business Process Improvement methodology and tools available.

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The eventual corrective action may involve modifications to the machine design or configuration, modifications to the operation or application of the machine, modifications to machine maintenance practices, additional manpower, training, or other support resources. It may become necessary to involve the additional resource from other groups within the organization or from Caterpillar.

4.0 Benefits

Successful implementation and execution of a Continuous Improvement program will:

- Deliver quality support services in an efficient manner
- Improve profitability
- Improve customer satisfaction through
 - o Increased availability
 - o Lower cost per ton
 - o Increased productivity

5.0 Resources Required

Human Resources: Human resource requirements associated with the implementation and execution of Continuous Improvement include:

- Team coordinator
 - o Provides overall leadership to team.
- Subject Matter Experts
 - o Function as team members with specific areas of expertise

Logistical Resources: Logistical resource requirements associated with the implementation and execution of Continuous Improvement include:

- Information
 - o Regular input (monthly reports) from other functional areas
 - o KPI's/Top Tier Metrics
 - o Problem reports, etc.

6.0 Supporting Attachments / References

- 6 Sigma or similar Business Process Improvement methodology & training

See the attached files:

- Cat Global Mining Equipment Management Process Map
- “Metrics (KPI's) to Assess Process Performance” – PDF Document
- Performance Metrics for Mobile Mining Equipment



7.0 Related Best Practices

1007-2.0-1109

Performance Evaluation (Strategic)

will be published Jan 2008

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8.0 Acknowledgements

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